

ANSWERS

1.

- (a) (i) R B1
- (ii) $0.5R$ B1
- (iii) $2.5R$... (allow e.c.f. from (ii)) B1
- (b) (i) $I_1 + I_2 = I_3$ B1
- (ii) $E_2 = I_3 R + I_2 R$ B1
- (iii) $E_1 - E_2 = 2I_1 R - I_2 R$ B1

2.

- (a) energy transferred from source / changed from some form to electrical M1
 per unit charge (to drive charge round a complete circuit) A1
- (b) and power in $R = I^2 X$ M1
 $E = I(X + r)$ M1
 power in cell = EI and algebra clear leading to ratio = $X / (X + r)$ A1
- (c) (i) 1.4 W A1
 0.40Ω (allow $\pm 0.05 \Omega$) A1
- (ii) current in circuit = $\sqrt{1.4/0.4} = 1.87 \text{ A}$ C1
 $1.5 = 1.87(r + 0.40)$ C1
 $r = 0.40 \Omega$ A1
- (d) *either* less power lost / energy wasted / lost
or greater efficiency (of energy transfer) B1

3.

- (a) total resistance in series = $2R$ M1
 total resistance in parallel = $\frac{1}{2}R$ A0
 ratio is $2R / \frac{1}{2}R = 4$ (allow mark if clear numbers in the ratio)A0

- (b) at 1.5 V, current is 0.10 AC1
 resistance = $V/I = \frac{1.5}{0.1}$
 = 15Ω A1
 (use of tangent or any other current scores no marks)

(c)

	p.d. across each lamp / V	resistance of each lamp / Ω	combined resistance / Ω
series	1.5	15	30
parallel	3.0	20	10

column 1A1
 columns 2 and 3: max 3 marks with -1 mark for each error or omissionA3

- (d) (i) ratio is 3(allow e.c.f.)A1
 (ii) resistance increases as potential difference increasesB1
 increasing p.d. increases currentB1
 current increases non-linearly so resistance increasesB1

4.

- (a) either $P \propto V^2$ or $P = V^2/R$ C1
 reduction = $(230^2 - 220^2)/230^2$
 = 8.5 %A1

- (b) (i) zeroA1
 (ii) 0.3(0)AA1

- (c) (i) correct plots to within ± 1 mmB1
 (ii) reasonable line/curve through points giving current as 0.12A
 allow $\pm 0.005A$ B1

- (iii) $V = IR$ C1
 $V = 0.12 \times 5.0$
 = 0.6(0)VA1

- (d) circuit acts as a potential divider/current divides/current in AC not the same as current in BC B1
 resistance between A and C not equal to resistance between C and B B1
 or current in wire AC $\times R$ is not equal to current in wire BC $\times R$ B1
 any 2 statements

5.

- (a) (i) either $P = V^2 / R$ or $P = VI$ and $V = IR$ C1
 $R = 4.0 \Omega$ A1

- (ii) sketch vertical axis labelled appropriately B1
 (straight) line from origin then curved in correct direction B1
 line passes through 12 V, 3.0 A B1

- (b) (i) 2.0 kW A1

- (ii) 0.5 kW A1

- (iii) total resistance = $3R / 2$ C1
 power = 0.67 kW A1

6.

- (a) (i) $I = 12 / (6 + 12)$ C1
 minimum current = 0.67 A A1

- (ii) correct start and finish points M1
 correct shape for curve with decreasing gradient A1

- (b) maximum current = 2.0 A A1
 minimum current = 0 A1

- (c) (i) smooth curve starting at (0,0) with decreasing gradient M1
 end section not horizontal A1

- (ii) full range of current / p.d. possible
 or currents / p.d. down to zero
 or brightness ranging from off to full brightness B1

7.

(a) (i) energy converted from chemical to electrical when charge flows through cell or round complete circuit B1

(ii) (resistance of the cell) causing loss of voltage or energy loss in cell B1

(b) (i) $E_B - E_A = I(R + r_B + r_A)$
 $12 - 3 = I(3.3 + 0.1 + 0.2)$ C1
 $I = 2.5 \text{ A}$ A1

(ii) Power = $E \times I$
 $= 12 \times 2.5$ C1
 $= 30 \text{ W}$ A1

(iii) $P = I^2 \times R$ or $P = V^2 / R$ or $P = VI$
 $= (2.5)^2 \times 3$ $= 9^2 / 3.6$ $= 9 \times 2.5$ C1
 $= 22.5 \text{ J s}^{-1}$ A1

(c) power supplied from cell B is greater than energy lost per second in circuit B1

8.

(a) (i) $I_1 + I_3 = I_2$ A1

(ii) $E_1 = I_2 \frac{R_2}{2} + I_1 \frac{R_2}{2} + I_1 R_1 + I_1 r_1$ A1

(iii) $E_1 - E_2$ B1
 $= -I_3 r_2 + I_1 (R_1 + r_1 + R_2 / 2)$ B1

(b) p.d. across BJ of wire changes / resistance of BJ changes B1
there is a difference in p.d across wire and p.d. across cell E_2 B1

9.

(a) (i) sum of e.m.f.'s = sum of p.d.'s around a loop/circuit B1

(ii) energy B1

(b) (i) $2.0 = I \times (4.0 + 2.5 + 0.5)$ C1
 $I = 0.286 \text{ A}$ (allow 2 s.f.) A1
(If total resistance is not 7Ω , 0/2 marks)

(ii) $R = [0.90 / 1.0] \times 4 (= 3.6)$ C1
 $V = I R = 0.286 \times 3.6 = 1.03 \text{ V}$ A1
(If factor of 0.9 not used, then 0/2 marks)

- (iii) $E = 1.03\text{ V}$ A1
- (iv) *either* no current through cell B
or p.d. across r is zero B1

10.

- (a) total resistance = $20\text{ (k}\Omega\text{)}$ C1
 current = $12 / 20\text{ (mA)}$ or potential divider formula C1
 p.d. = $[12 / 20] \times 12 = 7.2\text{ V}$ A1
- (b) parallel resistance = $3\text{ (k}\Omega\text{)}$ C1
 total resistance $8 + 3 = 11\text{ (k}\Omega\text{)}$ C1
 current = $12 / 11 \times 10^3 = 1.09 \times 10^{-3}$ or $1.1 \times 10^{-3}\text{ A}$ A1
- (c) (i) LDR resistance decreases M1
 total resistance (of circuit) is less hence current increases A1
- (ii) resistance across XY is less M1
 less proportion of 12 V across XY hence p.d. is less A1

11.

- (a) (i) sum of currents into a junction = sum of currents out of junction B1
- (ii) charge B1
- (b) (i) $\Sigma E = \Sigma IR$
 $20 - 12 = 2.0(0.6 + R)$ (not used 3 resistors 0/2) C1
 $R = 3.4\ \Omega$ A1
- (ii) $P = EI$ C1
 $= 20 \times 2$
 $= 40\text{ W}$ A1
- (iii) $P = I^2 R$ C1
 $P = (2)^2 \times (0.1 + 0.5 + 3.4)$
 $= 16\text{ W}$ A1
- (iv) efficiency = useful power / output power C1
 $24 / 40 = 0.6$ or $12 \times 2 / 20 \times 2$ or 60% A1

12.

- (a) (i) $I_1 = I_2 + I_3$ B1
- (ii) $I = V / R$ or $I_2 = 12 / 10$ (= 1.2 A) C1
 $R = [1/6 + 1 / 10]^{-1}$ [total $R = 3.75 \Omega$] or $I_3 = 12 / 6$ (= 2.0 A) C1
 $I_1 = 12 / 3.75 = 3.2 \text{ A}$ or $I_1 = 1.2 + 2.0 = 3.2 \text{ A}$ A1
- (iii) power = VI or I^2R or V^2 / R C1
- $$x = \frac{\text{power in wire}}{\text{power in series resistors}} = \frac{I_2^2 R_w}{I_3^2 R_s} \text{ or } \frac{VI_2}{VI_3} \text{ or } \frac{V^2 / R_w}{V^2 / R_s}$$
- C1
- $$x = 12 \times 1.2 / 12 \times 2.0 = 0.6(0) \text{ allow } 3 / 5 \text{ or } 3:5$$
- A1
- (b) p.d. BC: $12 - 12 \times 0.4 = 7.2 \text{ (V)}$ / p.d. AC = 4.8 (V) C1
 p.d. BD: $12 - 12 \times 4 / 6 = 4.0 \text{ (V)}$ / p.d. AD = 8.0 (V) C1
 p.d. = 3.2 V A1

13.

- (a) e.m.f. = chemical energy to electrical energy M1
 p.d. = electrical energy to thermal energy M1
 idea of per unit charge A1
- (b) $E = I(R + r)$ or $I = E / (R + r)$ (any subject) B1
- (c) (i) $E = 5.8 \text{ V}$ B1
- (ii) evidence of gradient calculation or calculation with values from graph
 e.g. $5.8 = 4 + 1.0 \times r$ C1
 $r = 1.8 \Omega$ A1
- (d) (i) $P = VI$ C1
 $P = 2.9 \times 1.6 = 4.6$ (4.64) W A1
- (ii) power from battery = $1.6 \times 5.8 = 9.28$ or efficiency = VI / EI C1
 efficiency = $(4.64 / 9.28) \times 100 = 50 \%$ or $(2.9 / 5.8) \times 100 = 50\%$ A1

14.

(a) p.d. = $\frac{\text{work done} / \text{energy transformed}}{\text{charge}}$ (from electrical to other forms)

B1

(b) (i) maximum 20 V

A1

(ii) minimum = $(600 / 1000) \times 20$
= 12 V

C1

A1

(c) (i) use of 1.2 k Ω

M1

$$1/1200 + 1/600 = 1/R, R = 400 \Omega$$

A1

(ii) total parallel resistance ($R_2 + \text{LDR}$) is less than R_2
(minimum) p.d. is reduced

M1

A1

15.

(a) there are no lost volts/energy lost in the battery
or there are no lost volts/energy lost in the internal resistance

B1

(b) the current/ I decreases (as R increases)

M1

p.d. decreases (as R increases)

A1

or

the parallel resistance (of X and R) increases

M1

p.d. across parallel resistors increases, so p.d. (across Y) decreases

A1

(c) (i) current = 2.4 (A)

C1

$$\text{p.d. across AB} = 24 - 2.4 \times 6 = 9.6 \text{ V}$$

M1

or

$$\text{total resistance} = 10 \Omega \quad (= 24 \text{ V} / 2.4 \text{ A})$$

C1

$$(\text{parallel resistance} = 4 \Omega), \quad \text{p.d.} = 24 \times (4 / 10) = 9.6 \text{ V}$$

M1

(ii) $R(\text{AB}) = 9.6 / 2.4 = 4.0 \Omega$

C1

$$1/6 + 1/X = 1/4 \quad [\text{must correctly substitute for } R]$$

C1

$$X = 12 \Omega$$

A1

or

$$I_R = 9.6 / 6.0 = 1.6 \text{ (A)}$$

(C1)

$$I_X = 2.4 - 1.6 = 0.8 \text{ (A)}$$

(C1)

$$X (= 9.6 / 0.8) = 12 \Omega$$

(A1)

(iii) power = VI or EI or V^2/R or E^2/R or I^2R C1
 $= 24 \times 2.4$ or $(24)^2/10$ or $(2.4)^2 \times 10$
 $= 57.6 \text{ W}$ (allow 2 or more s.f.) A1

(d) power decreases M0

e.m.f. constant or power = $24 \times$ current, and current decreases
 or e.m.f. constant or power = $24^2/\text{resistance}$, and resistance increases A1

16.

(a) (i) in series $2X$ or in parallel $X/2$ M1
 other relationship given and $4\times$ greater in series (than in parallel) A1

(ii) due to the internal resistance B1

total resistance for series circuit is not four times greater than resistance
 for parallel circuit B1

(iii) 1. $E = I_1(2X + r)$ or $12 = 1.2(2X + r)$ A1

2. $E = I_2(X/2 + r)$ or $12 = 3.0(X/2 + r)$ A1

(iv) $2X + r = 10$ and $X/2 + r = 4$
 $X = 4.0 \Omega$ A1

(b) $P = I^2R$ or V^2/R or VI C1

ratio = $[(1.2)^2 \times 4] / [(1.5)^2 \times 4]$
 $= 0.64$ A1

(c) the resistance (of a lamp) changes with V or I B1

V or I is greater in parallel circuit or circuit 2
 or V or I is less in series circuit or circuit 1 B1

17.

(a) $R = \rho l / A$ C1

$A = [\pi \times (0.38 \times 10^{-3})^2] / 4$ ($= 0.113 \times 10^{-6} \text{ m}^2$) C1

$R = (4.5 \times 10^{-7} \times 1.00) / ([\pi \times (0.38 \times 10^{-3})^2] / 4) = 4.0 \text{ (3.97)} \Omega$ M1

(b) (i) $I = V/R$ $= 2.0 / 5.0 = 0.4(0) A$	C1 A1
(ii) p.d. across BD $= 4 \times 0.4 = 1.6 V$	A1
(iii) p.d. across BC (l) $= 1.5 (V)$	C1
BC (l) $= (1.5 / 1.6) \times 100 = 94 (93.75) \text{ cm}$	A1
(c) p.d. across wire not balancing e.m.f. of cell OR cell Y has current energy lost or lost volts due to internal resistance	B1 B1
18.	
(a) curved line showing decreasing gradient with temperature rise	M1
smooth line not touching temperature axis, not horizontal or vertical anywhere	A1
(b) (i) (no energy lost in battery because) no/negligible internal resistance	B1
(ii) $I = V/R$	
$= 8/15 \times 10^3 \text{ or } 1.6/3.0 \times 10^3 \text{ or } 2.4/4.5 \times 10^3 \text{ or } 12/22.5 \times 10^3$	C1
$= 0.53 \times 10^{-3} A$	A1
(iii) p.d. across X $= 12 - 8.0 - 3.0 \times 10^3 \times 0.53 \times 10^{-3} (= 2.4 V)$	C1
$R_X = 2.4 / (0.53 \times 10^{-3})$	C1
or	
$R_{\text{tot}} = 12 / 0.53 \times 10^{-3} (= 22.5 \times 10^3 \Omega)$	(C1)
$R_X = (22.5 - 15.0 - 3.0) \times 10^3$	(C1)
$4.5(2) \times 10^3 \Omega$	A1
(iv) resistance decreases hence current (in circuit) is greater	M1
p.d. across X and Y is greater hence p.d across Z decreases	A1
or explanation in terms of potential divider:	
R_Z decreases so $R_Z / (R_X + R_Y + R_Z)$ is less	(M1)
therefore p.d. across Z decreases	(A1)

19.

(a) $R = \rho l / A$ C1

$$= (5.1 \times 10^{-7} \times 0.50) / \pi(0.18 \times 10^{-3})^2 = 2.5 \text{ (2.51)} \Omega$$
 M1

(b) (i) resistance of CD = $8 \times$ resistance of AB = 20Ω C1

$$\text{circuit resistance} = [1/5.0 + 1/20]^{-1} = 4.0 \Omega$$
 C1

$$\text{current} = V/R = 6.0/4.0$$
 C1

$$= 1.5 \text{ A}$$
 A1

(ii) power in AB = $I^2 R$ or power = V^2 / R C1

$$= (1.2)^2 \times 2.5 = 3.6 \text{ W} \quad \quad \quad = (3.0)^2 / 2.5 = 3.6 \text{ W}$$
 A1

(iii) potential drop A to M = $1.25 \times 1.2 = 1.5 \text{ V}$ M1

$$\text{potential drop C to N} = 3.0 \text{ V}$$

$$\text{p.d. MN} = 1.5 \text{ V}$$
 A1

20.

(a) $\frac{\text{work done or energy (transformed) (from electrical to other forms)}}{\text{charge}}$ B1

(b) (i) 1. $V = IR$ or $E = IR$ C1

$$I = 14/6.0$$

$$= 2.3 \text{ (2.33) A}$$
 A1

2. total resistance of parallel resistors = 8.0Ω C1

$$\text{current} = 14/(6.0 + 8.0)$$

$$= 1.0 \text{ A}$$
 A1

(ii) $P = EI$ (allow $P = VI$) or $P = V^2 / R$ or $P = I^2 R$ C1

$$\text{change in power} = (14 \times 2.33) - (14 \times 1.0)$$

$$\text{or } (14^2 / 6.0) - (14^2 / 14)$$

$$\text{or } (2.33^2 \times 6.0) - (1.0^2 \times 14)$$

$$= 19 \text{ W (18 W if 2.3 A used)}$$
 A1

(c) $I = Anvq$

$$\text{ratio} = (0.50n/n) \times (1.8A/A) \quad \text{or} \quad \text{ratio} = 0.50 \times 1.8$$

$$= 0.90$$

21.

(a) total/sum of electromotive forces or e.m.f.s

= total/sum of potential differences or p.d.s

around a loop/(closed) circuit

C1

A1

M1

A1

(b) (i) (current in battery =) current in A + current in B or $I_A + I_B$

$$(I =) 0.14 + 0.26 = 0.40 \text{ A}$$

C1

A1

(ii) $E = V + Ir$

$$6.8 = 6.0 + 0.40r \quad \text{or} \quad 6.8 = 0.40(15 + r)$$

$r = 2.0 \, \Omega$

(iii) $R = V/I$

$$\text{ratio} (= R_A/R_B) = (6.0/0.14)/(6.0/0.26) \\ = 42.9/23.1 \text{ or } 0.26/0.14$$

$$= 1.9 \text{ (1.86)}$$

(iv) 1. $P = EI$ or VI or $P = I^2R$ or $P = V^2/R$

$$= 6.8 \times 0.40$$

$$= 0.40^2 \times 17$$

$$= 6.8^2 / 17$$

$$= 2.7 \text{ W } (2.72 \text{ W})$$

2. output power = VI

$$= 6.0 \times 0.40 (= 2.40 \text{ W})$$

$$\text{efficiency} = (6.0 \times 0.40) / (6.8 \times 0.40) = 2.40 / 2.72$$
$$= 0.88 \text{ or } 88\% \text{ (allow } 0.89 \text{ or } 89\%)$$

C1

A1

C1

A1

C1

A1

C1

A1

22.

6(a)	volt / ampere	B1
6(b)(i)	$R_T = [1/3.0 + 1/6.0]^{-1} + 4.0 (= 6.0 \Omega)$	C1
	$I = 1.5 / 6.0$	C1
	$= 0.25 \text{ A}$	A1
6(b)(ii)	$V_B = 0.5 \text{ V}$	A1
	$I = 0.5 / 3.0$	
	$= 0.17 (0.167) \text{ A}$	
6(b)(iii)	$P = I^2 R \text{ or } VI \text{ or } V^2 / R$	C1
	ratio $= (0.167^2 \times 3.0) / (0.25^2 \times 4.0)$	A1
	$= 0.33$	
6(c)(i)	vary/change/different radius/diameter/ <u>cross-sectional</u> area (of wire)	B1
6(c)(ii)	$v = I / Ane$	C1
	ratio $= \frac{(I_B / A_B)}{(I_C / A_C)} \text{ or } \frac{I_B \times A_C}{I_C \times A_B}$	
	$(R \propto 1 / A \text{ so}) \text{ ratio} = \frac{I_B \times R_B}{I_C \times R_C} = \frac{0.167 \times 3.0}{0.25 \times 4.0}$ $= 0.50$	A1
6(d)(i)	0.25 A to 0.13 (0.125) A or halved	A1
6(d)(ii)	no change	A1

23.

7(a)	energy transformed from <u>chemical to electrical</u> / unit charge (driven around a complete circuit)	B1
7(b)(i)	the current decreases (as resistance of Y increases)	M1
	lost volts go down (as resistance of Y increases)	M1
	p.d. AB increases (as resistance of Y increases)	A1
7(b)(ii)1.	$1.50 = 0.180 \times (6.00 + 0.200 + R_X)$	C1
	$R_X = 2.1(3) \Omega$	A1
7(b)(ii)2.	p.d. AB $= 1.5 - (0.180 \times 0.200) \text{ or } 0.18 \times (2.13 + 6.00)$	C1
	$= 1.46(4) \text{ V}$	A1
7(b)(ii)3.	efficiency $= (\text{useful}) \text{ power output} / (\text{total}) \text{ power input or } IV / IE$	C1
	$(= 1.46 / 1.5) = 0.97 [0.98 \text{ if full figures used}]$	A1

24.

5(a)	$I_1 + I_2 = I_3$ [any subject]	B1
5(b)	$E_1 + E_3 = I_1 R_1 + I_3 R_3 + I_3 R_4$ [any subject]	B1
5(c)	$E_1 - E_2 = I_1 R_1 - I_2 R_2$ [any subject]	B1

25.

6(a)(i)	<u>sum of</u> current(s) into junction = <u>sum of</u> current(s) out of junction or (algebraic) sum of current(s) at a junction is zero	B1
6(a)(ii)	charge	B1
6(b)(i)1.	$E = I^2 R t$ or $E = V I t$ or $E = (V^2 / R) t$	C1
	$E = 2.5^2 \times 2.0 \times 5.0 \times 60$ or $5.0 \times 2.5 \times 5.0 \times 60$ or $(5.0^2 / 2.0) \times 5.0 \times 60$ $= 3800 \text{ J}$	A1
6(b)(i)2.	p.d. = $8.0 - (2.0 \times 2.5)$ $= 3.0 \text{ V}$	A1
6(b)(ii)	$I_X = 3.0 / 15 = 0.20 \text{ (A)}$	C1
	$I_Y = 2.5 - 0.20 = 2.3 \text{ (A)}$	C1
	$R_Y = 3.0 / 2.3$ $= 1.3 \Omega$	A1
	or	
	$R_T = 3.0 / 2.5 = 1.2 (\Omega)$ or $(8.0 / 2.5) - 2.0 = 1.2 (\Omega)$	(C1)
	$1 / 1.2 = 1 / 15 + 1 / R_Y$	(C1)
	$R_Y = 1.3 \Omega$	(A1)
6(b)(iii)1.	Z has larger radius/diameter/(cross-sectional) area	B1
	Z has (material of) smaller resistivity/greater conductivity	B1
6(b)(iii)2.	current/ I (in battery) increases	M1
	($P = EI$ so) power/ P (produced by battery) increases	A1

26.

6(a)	<u>sum of</u> e.m.f.(s) equal to <u>sum of</u> p.d.(s) around a loop/around a closed circuit	M1
6(b)(i)	current in variable resistor = $(6.0 / 2.4) + (6.0 / 1.2)$ (= 7.5 A)	C1
	p.d. across variable resistor = $9.0 - 6.0$ (= 3.0 V)	C1
	$R = 3.0 / 7.5$ $= 0.40 \Omega$	A1
	or	
	$\frac{1}{R_T} = \frac{1}{2.4} + \frac{1}{1.2}$ $R_T = 0.80 (\Omega)$	(C1)
	$\frac{3}{9} = \frac{R}{(R + 0.80)}$ or $\frac{3}{R} = \frac{6}{0.8}$	(C1)
	$R = 0.40 \Omega$	(A1)
6(b)(ii)	$P = V^2 / R$ or $P = I^2 R$ or $P = IV$	C1
	$P = 6.0^2 / 24$ or $2.5^2 \times 2.4$ or 6.0×2.5 $= 15 \text{ W}$	A1

6(b)(iii)	1. $R = \frac{\rho L}{A}$	C1
	ratio = $(2.4/1.2) \times (3/1)$ = 6.0	A1
	2. $(I = nAvq)$ $I_X/I_Y = 2.5/5.0$ or $1.2/2.4$ or 0.5	C1
	ratio = $(2.5/5.0) \times (1/3)$ or $(1.2/2.4) \times (1/3)$ = 0.17	A1

27.

6(a)	joule/coulomb	B1
6(b)(i)	$7.0 = (I \times 5.2) + (I \times 6.0) + 1.4$	C1
	$I = 0.50 \text{ A}$	A1
6(b)(ii)	$R = 1.4/0.50$ = 2.8Ω	A1
6(b)(iii)	$P = EI$ or $P = VI$ or $P = I^2 R$ or $P = V^2/R$	C1
	efficiency = $[(0.50^2 \times 6.0)/(7.0 \times 0.50)] (\times 100)$ or efficiency = $[(0.50 \times 3.0)/(7.0 \times 0.50)] (\times 100)$ or efficiency = $[(3.0^2/6.0)/(7.0 \times 0.50)] (\times 100)$	C1
	efficiency = 43%	A1
6(b)(iv)	$R = \rho l/A$	C1
	$\alpha = \rho/R$	A1
	= $3.7 \times 10^{-7}/6.0$	
	= $6.2 \times 10^{-8} \text{ m}$	

28.

7(a)	<u>sum of</u> current(s) in(to) junction = <u>sum of</u> current(s) out of junction or (algebraic) sum of current(s) at a junction is zero	B1
7(b)(i)	1. potential difference = 0	A1
	2. potential difference = 9.6 V	A1
7(b)(ii)	for resistance in parallel: $(1/R_T) = (1/400) + (1/400)$ $R_T = 200 (\Omega)$	C1
	$V/9.6 = 200/600$	C1
	$V = 3.2 \text{ V}$	A1

29.

6(a)	energy is dissipated in the internal resistance	B1
6(b)	$E = V + Ir$	B1
6(c)(i)	(graph shows) maximum value of potential difference is 2.8 (V) or (graph shows) when current/ I (from battery) is zero, V is 2.8 (V) / E	B1
6(c)(ii)	$r = (-)\text{gradient}$ or $r = (E - V) / I$ or substituted values from the graph for E , V and I	C1
	$r = 1.4 \Omega$	A1
6(d)(i)	$R = 2.1 / 0.50$ $= 4.2 \Omega$	A1
6(d)(ii)	number $= 0.50 / 1.60 \times 10^{-19}$ $= 3.1 \times 10^{18}$	A1
6(d)(iii)	energy $= EIt$ or $P = EI$ and $P = W / t$	C1
	$(9.2 - 1.6) \times 10^3 = 2.8 \times 0.50 \times t$	C1
	$t = 5.4 \times 10^3 \text{ s}$	A1

30.

5(a)	<u>sum of</u> e.m.f.(s) = <u>sum of</u> p.d.(s) around a loop/around a closed circuit	M1
		A1
5(b)(i)	1. $1/R = 1/R_1 + 1/R_2$ $1/R = 1/90 + 1/18$ $R = 15 \Omega$	C1
		A1
	2. $I = V/R$	C1
	$I = 4.8/15$ or $I = 4.8/90 + 4.8/18$ $I = 0.32 \text{ A}$	A1
5(b)(ii)	$E = V + Ir$ or $E = I(R + r)$	C1
	$5.6 = 4.8 + 0.32 r$ so $r = 2.5 (\Omega)$ or $5.6 = 0.32 \times (15 + r)$ so $r = 2.5 (\Omega)$	A1
5(b)(iii)	$P = EI$ or $P = VI$ or $P = I^2 R$ or $P = V^2 / R$	C1
	ratio $= (0.32^2 \times 2.5) / (5.6 \times 0.32)$ or $0.256 / 1.792$	C1
	$= 0.14$	A1
5(c)	$7.2 - 5.6 - 2.5I - 3.5I = 0$	C1
	$I = 0.27 \text{ A}$	A1

31.

6(a)	work done / charge or energy (transferred from electrical to other forms) / charge	B1
6(b)	for $V < 0.25 \text{ V}$ resistance is infinite/very high (as current is zero)	B1
	for $V > 0.25 \text{ V}$ resistance decreases (as V increases)	B1
6(c)(i)	$R = V / I$	C1
	$= 0.75 / (15 \times 10^{-3})$	C1
	$= 50 \Omega$	A1
6(c)(ii)	1. $V_Y = 15 \times 10^{-3} \times 60 (= 0.90 \text{ V})$	C1
	$V_X = 2.0 - 0.90 - 0.75 (= 0.35 \text{ V})$	C1
	$R_X = 0.35 / (15 \times 10^{-3})$	A1
	$= 23 \Omega$	
	or	
	total $R = 60 + 50 + R_X$	(C1)
	$60 + 50 + R_X = 2.0 / (15 \times 10^{-3})$	(C1)
	$R_X = 23 \Omega$	(A1)
	2. $P = VI$ or $P = EI$ or $P = I^2R$ or $P = V^2 / R$	C1
	ratio $= \frac{(15 \times 10^{-3})^2 \times 60}{2.0 \times 15 \times 10^{-3}}$ or $\frac{0.90 \times 15 \times 10^{-3}}{2.0 \times 15 \times 10^{-3}}$ or $\frac{(0.90^2 / 60)}{2.0 \times 15 \times 10^{-3}}$	A1
	$= 0.45$	

32.

6(a)	<u>sum of</u> current(s) into junction = <u>sum of</u> current(s) out of junction or (algebraic) sum of current(s) at a junction is zero	B1
6(b)(i)	$R = V / I$	C1
	$= 0.60 / 7.5 \times 10^{-3}$	C1
	$= 80 \Omega$	A1
6(b)(ii)	resistance decreases	B1
6(c)(i)	$E = 0.60 + 0.30$ $= 0.90 \text{ V}$	A1
6(c)(ii)	$(I =) 9.3 - 7.5$	C1
	$I = 1.8 \text{ (mA)}$ or $1.8 \times 10^{-3} \text{ (A)}$	A1
	$R = 0.90 / 1.8 \times 10^{-3}$	
	$= 500 \Omega$	
	or	
	total resistance $= 0.90 / 9.3 \times 10^{-3} = 96.8 \text{ (}\Omega\text{)}$	(C1)
	total resistance of diode and X $= 0.90 / 7.5 \times 10^{-3} = 120 \text{ (}\Omega\text{)}$	
	$1 / 96.8 = 1 / R + 1 / 120$	
	$R = 500 \Omega$	(A1)

6(c)(iii)	$P = VI$ or I^2R or V^2/R	C1
	$= 0.60 \times 7.5 \times 10^{-3}$ or $(7.5 \times 10^{-3})^2 \times 80$ or $0.60^2 / 80$	A1
	$= 4.5 \times 10^{-3} \text{ W}$	
6(c)(iv)	current = 2.5 mA	A1

33.

6(a)(i)	$R = V/I$	C1
	resistance = $(12 / 0.20) / 2$ or $6 / 0.20$	A1
	$= 30 \Omega$	
6(a)(ii)	$I = 0.50 - 0.20$ (= 0.30 A)	C1
	$R + 28 = 12 / 0.30$ (= 40 Ω)	A1
	$R = 12 \Omega$	
6(b)	p.d. across lamp = 0.20×30 (= 6.0 V)	C1
	p.d. across $R = 0.30 \times 12$ (= 3.6 V)	C1
	$V_{XY} = 6.0 - 3.6$	A1
	$= 2.4 \text{ V}$	
	or	
	p.d. across lamp = 0.20×30 (= 6.0 V)	(C1)
	p.d. across 28 Ω resistor = 0.30×28 (= 8.4 V)	(C1)
6(c)	$P = VI$ or $P = EI$ or $P = I^2R$ or $P = V^2/R$	C1
	ratio = $(6.0 \times 0.20) \times 2 / (12 \times 0.50)$ or $0.20 / 0.50$	A1
	$= 0.40$	
6(d)	no change to V across lamps, so power in lamps unchanged or current in battery/total current increases (and e.m.f. the same) so power produced by battery increases	B1
	both the above statements and so the ratio decreases	B1

34.

5(a)(i)	$R = \rho L / A$	C1
	$A = (2.6 \times 10^{-8} \times 59) / 3.4 = 4.5 \times 10^{-7} \text{ m}^2$	A1
5(a)(ii)	$I = 1.8 / 3.4$ $= 0.53 \text{ A}$	A1
5(a)(iii)	$I = Anvq$ $v = 0.53 / (4.5 \times 10^{-7} \times 6.1 \times 10^{28} \times 1.60 \times 10^{-19})$	C1
	$= 1.2 \times 10^{-4} \text{ m s}^{-1}$	A1
5(b)(i)	(cross-sectional) area/A is less	M1
	(I , n , e the same so) average drift speed is greater	A1
5(b)(ii)	(area is less so) more resistance/ R	M1
	(I is the same, so) more power/ P	A1
	or	
	($P = I^2 \rho L / A$ so) $P \propto 1 / A$	(M1)
	(A is less so) more P	(A1)
5(c)(i)	180 Ω and 90 Ω resistors shown connected in parallel	B1
5(c)(ii)	resistors connected in parallel labelled as 180 Ω and 90 Ω and the other resistor labelled as 30 Ω	M1
	V_{OUT} or 8.0 V labelled across the two resistors in parallel	A1

35.

6(a)(i)	energy is dissipated in the internal resistance/ r	B1
6(a)(ii)	1. $I = Q / t$	C1
	$= 750 / 1500$	A1
	$= 0.50 \text{ A}$	
	2. $V = W / Q$ or $V = W / It$	C1
	$= 5700 / 750$ or $5700 / (0.50 \times 1500)$	A1
	$= 7.6 \text{ V}$	
	or	
	$V = P / I$ and $P = W / t$	(C1)
	$V = 3.8 / 0.50$	(A1)
	$= 7.6 \text{ V}$	
6(b)(i)	90 Ω and 45 Ω resistors shown connected in parallel	B1
	the resistors connected in parallel labelled as 90 Ω and 45 Ω with the other resistor labelled as 20 Ω	M1
	V_{OUT} or 3.6 V labelled across the 20 Ω resistor	A1

36.

7(a)	volt / ampere	B1
7(b)	$R = \rho L / A$	C1
	$A = 460 \times 10^{-9} \times 2.5 / 3.2$	C1
	$= 3.6 \times 10^{-7} \text{ m}^2$	A1
7(c)(i)	energy is dissipated in the internal resistance/ r	B1
7(c)(ii)	$E = IR + Ir$ or $E = I(R + r)$	B1
7(c)(iii)	$P = I^2 R$ or $P = I^2 r$	C1
	$I = E / 2r$	A1
	(so) $P = E^2 / 4r$	

37.

6(a)	<u>work (done)/energy (transferred from electrical to other forms)</u> charge	B1
6(b)	$R = \rho L / A$	B1
	$V = LA$ and (so) $R = \rho V / A^2$ (with ρ and V constant)	B1
6(c)	$E = IR + Ir$ or $E = I(R + r)$ or $E - Ir = IR$ and $R = (E / I) - r$	A1
6(d)(i)	$P = I^2 R$ or $P = IV$ or $P = V^2 / R$	C1
	$R = 5.4 \text{ } (\Omega)$ or $V = 10.8 \text{ (V)}$	C1
	$P = 2.0^2 \times 5.4$	A1
	$= 22 \text{ W}$	
6(d)(ii)	1. $r = 0.60 \text{ } \Omega$	A1
	2. $E = \text{gradient}$	C1
	$= \text{e.g. } 5.4 / 0.45$	A1
	$= 12 \text{ V}$	

38.

6(a)	<u>sum of current(s) into junction = sum of current(s) out of junction</u> or (algebraic) sum of current(s) at a junction is zero	B1
6(b)(i)	$I = 3.6 - 2.1$ $= 1.5$	C1
	$V = 4.4$	C1
	$R = 4.4 / 1.5$ $= 2.9 \text{ } \Omega$	A1
6(b)(ii)	$12.0 = 4.4 + 3.6r$ or $12.0 = 3.6(1.2 + r)$	C1
	$r = 2.1 \text{ } \Omega$	A1
6(b)(iii)	$t = (470 \times 10^3 - 240 \times 10^3) / (12 \times 3.6)$	C1
	$= 5300 \text{ s}$	A1
6(b)(iv)	$I = Anvq$	C1
	ratio = $(360A / A) \times (2.5n / n)$ or 360×2.5	
	$= 900$	A1

39.

5(a)	sum of e.m.f.(s) = sum of p.d.(s) or (algebraic) sum of e.m.f.(s) and p.d.(s) is zero	M1
	around a loop/around a closed circuit	A1
5(b)(i)	$I = 1.8 / 0.90$ $= 2.0 \text{ A}$	A1
5(b)(ii)	$Q = It$	C1
	number = $(2.0 \times 45) / 1.60 \times 10^{-19}$ $= 5.6 \times 10^{20}$	A1
5(b)(iii)	$4.0 = 1.8 + [2.0 \times (0.35 + R)]$ or $4.0 = 2.0 \times (0.90 + 0.35 + R)$	C1
	$R = 0.75 \Omega$	A1
5(c)(i)	$1.2 / 1.8 = 0.30 / L$	C1
	$L = 0.45 \text{ m}$	A1
5(c)(ii)	p.d. across XY decreases/p.d. across XP decreases	B1
	(so) P is moved towards Y/away from X/to the right	B1

40.

5(a)	energy per unit charge	B1
	energy transferred by source driving charge around the complete circuit or energy transferred from other forms to electrical energy	B1
5(b)	there is a p.d. across the internal resistance/ r	B1
	change in current/ I results in a change in p.d. across the internal resistance	B1
	$V = E - \text{p.d. across internal resistance}$ or change in p.d. across r causes a change in V (as e.m.f. is constant)	B1
5(c)(i)	$E = 7.4 \text{ V}$	A1
5(c)(ii)	maximum current = 0.92 A	A1
5(c)(iii)	$r = E / I_{\text{MAX}}$ or (–)gradient	C1
	e.g. $r = 7.4 / 0.92$ $= 8.0 \Omega$	A1
5(d)	straight line with negative gradient that is smaller in magnitude than the original line	B1
	line which would have intercept on V -axis below the original line	B1

41.

5(a)	sum of current(s) in = sum of current(s) out or (algebraic) sum of current(s) is zero	M1
	at a junction (in a circuit)	A1
5(b)(i)	(current in R_4 or R_1 =) $0.30 + 0.30$ (= 0.60 A)	B1
	(R =) $2.4 / 0.60 = 4.0 \text{ } (\Omega)$	A1
	or	
	(p.d. across R_3 or R_2 =) $2.4 / 2$ (= 1.2 V)	(B1)
	(R =) $1.2 / 0.30 = 4.0 \text{ } (\Omega)$	(A1)
5(b)(ii)	$E = 2.4 + 2.4 + 1.2$	C1
	= 6.0 V	A1
	or	
	total resistance = $10 \text{ } (\Omega)$	(C1)
	$E = 10 \times 0.60$ = 6.0 V	(A1)
5(c)	total resistance increases	B1
	current decreases (in battery) so total power decreases	B1
5(d)	resistivity = RA / L	C1
	= $4.0 \times \pi \times (240 \times 10^{-6})^2 / 0.67$	C1
	= $1.1 \times 10^{-6} \text{ } \Omega \text{ m}$	A1

42.

6(a)	$R = \rho L / A$	C1
	(R =) $5(.0) \times 10^{-7} \times 2(.0) / 3.3 \times 10^{-7} = 3.0 \text{ } \Omega$	A1
6(b)(i)	$I = 1.2 / 3.0$ = 0.40 A	A1
6(b)(ii)	$r = (1.50 - 1.20) / 0.40$ or $1.50 / 0.40 - 3.0$	C1
	= $0.75 \text{ } \Omega$	A1
6(c)	$E / 1.20 = 1.4 / 2.0$	C1
	$E = 0.84 \text{ V}$	A1
	or	
	$R_{XP} = (1.4 / 2.0) \times 3.0$ (= $2.1 \text{ } \Omega$)	(C1)
	$E = 2.1 \times 0.40$	
6(d)	$E = 0.84 \text{ V}$	(A1)
	(second wire has) larger resistance/resistance increases	M1
	p.d. across XY is larger/increases (for second wire) or p.d. across the (second) wire is larger/increases	M1
	(so) length XP (for second wire) is shorter	A1

43.

6(a)	sum of current(s) into junction = sum of current(s) out of junction or (algebraic) sum of current(s) at a junction is zero	B1
6(b)(i)	same potential difference (across X and Y as in parallel)	B1
	power = V^2 / R (and $R_X > R_Y$) or power = VI and $I_X < I_Y$	M1
	(so) Y (dissipates more power)	A1
6(b)(ii)	$R = \rho L / A$ (and $R_X > R_Y$)	M1
	(so) Y (has the larger (cross-sectional) area)	A1
6(c)(i)	current = $5.4 / 1800$ $= 3.0 \times 10^{-3} \text{ A}$	A1
6(c)(ii)	$5.4 / 9.0 = 1800 / (1800 + R_L)$ or $R_L = (9.0 - 5.4) / 3.0 \times 10^{-3}$	C1
	$R_L = 1200 \Omega$	A1
6(c)(iii)	resistance of LDR / R_L decreases	B1
	current (in the circuit) increases (so) voltmeter reading increases	B1

44.

5(a)	current (through a conductor is directly) proportional to potential difference (across the conductor)	M1
	(provided that) temperature (of conductor remains) constant	A1
5(b)(i)	(ratio of) V / I increases (as p.d. increases)	B1
5(b)(ii)	(as p.d. increases, current increases so) temperature increases	B1
5(c)(i)	$I = 1.55 \text{ A}$	A1
5(c)(ii)	$P = VI$ or $P = I^2 R$ or $P = V^2 / R$	C1
	$= 6.0 \times 1.55 \times 2$ or $1.55^2 \times 3.87 \times 2$ or $(6.0^2 / 3.87) \times 2$	A1
	$= 19 \text{ W}$	
5(c)(iii)	$I = 1.78 - 1.55$ (= 0.23 A)	C1
	$R = 12.0 / 0.23$	A1
	$= 52 \Omega$	
5(d)	lamp P: p.d. across lamp decreases to zero so goes 'out'	B1
	lamp Q: p.d. across lamp increases to 12 V so gets brighter	B1

45.

5(a)	sum of e.m.f.(s) = sum of p.d.(s) or (algebraic) sum of e.m.f.(s) and p.d.(s) is zero	M1
	around a loop / around a <u>closed</u> circuit	A1
5(b)(i)	$1 / r_{(T)} = 1 / 0.59 + 1 / 0.59 + 1 / 0.59$	B1
	$(r_{(T)} =) 0.197 (\Omega)$	A1
	$(R =) 2.2 - 0.197 = 2.0 \Omega$	
	or	
	$I = 1.5 / 2.2 (= 0.68 \text{ A})$ and $i = 0.68 / 3$ (where I is the circuit current and i is the current from each cell)	(B1)
	$(E = IR + ir =) 1.5 = 0.68R + (0.68 / 3) \times 0.59$ and $R = 2.0 \Omega$	(A1)
5(b)(ii)	current = $1.5 / 2.2$	C1
	= 0.68 A	A1
	or	
	p.d. across cell = p.d. across conductor	(C1)
	$1.5 - 0.59I = 3I \times 2.0$ so $I = 0.228 \text{ A}$ (where I is current in cell)	
	current = 3×0.228	
	= 0.68 A	(A1)
	or	
	current in conductor = $3 \times$ current in cell	(C1)
5(c)(i)	$R = \rho L / A$	C1
	$R = 4\rho L / \pi d^2$	C1
	(ρ and L are the same so)	
	$R_A / R_B = 7.6^2 / 4.3^2$	
	= 3.1	A1
5(c)(ii)	$I = Anvq$ and I, n, q are same / equal / constant	B1
	$\frac{v_A}{v_B} = \frac{A_B}{A_A} = \frac{d_B^2}{d_A^2}$	A1
	ratio = $7.6^2 / 4.3^2$ = 3.1	
5(d)	combined internal resistance (of the cells) will be greater or total / circuit resistance (of circuit) greater (because a parallel resistance removed)	B1
	more 'lost volts' (inside each cell) or internal resistances take a greater share of total p.d. or conductor gets a smaller share of the total p.d. or current in <u>conductor</u> /total current decreases	M1
	(so) potential difference (across conductor) decreases	A1

46.

7(a)	$9.6 = 6.0 + (I \times 5800)$ or $3.6 = I \times 5800$	C1
	$I = 6.2 \times 10^{-4} \text{ A}$	A1
7(b)	$9.6 = 6.2 \times 10^{-4} \times (3400 + 5800 + R)$	C1
	or	
	$6.0 = 6.2 \times 10^{-4} \times (3400 + R)$	
	$R = 6.3 \times 10^3 \Omega$	A1
7(c)	$(\Delta E =) 9.6 \times 330 (= 3170 \text{ J})$	C1
	final stored energy = $2.6 \times 10^4 - 3170$ $= 2.3 \times 10^4 \text{ J}$	A1
7(d)(i)	decrease	B1
7(d)(ii)	decrease	B1
7(d)(iii)	increase	B1

47.

7(a)	$V / 9.0 = 1800 / (1800 + 1200)$	C1
	$V = 5.4 \text{ V}$	A1
	or	
	$I = 9.0 / (1800 + 1200) = 3.0 \times 10^{-3} \text{ (A)}$	(C1)
	$V = 3.0 \times 10^{-3} \times 1800$	
	$= 5.4 \text{ V}$	(A1)
7(b)(i)	$L / 1.2 = 5.4 / 9.0$ or $XZ / 1.2 = 5.4 / 9.0$	C1
	$L = 0.72 \text{ m}$	A1
	or	
	$L / 1.2 = 1800 / (1800 + 1200)$ or $XZ / 1.2 = 1.8 / (1.8 + 1.2)$	(C1)
	$L = 0.72 \text{ m}$	(A1)
7(b)(ii)	• (intensity) increase	B1
	• (power) increase	B1
	• (length XZ) decrease	B1